

SQL Views

20 · Views — Complete Reference Guide

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1. Create, Drop, Modify View

→ *Create a view summarising monthly sales*

```
CREATE VIEW Sales.V_Monthly_Summary AS
SELECT
    DATETRUNC(month, OrderDate) AS OrderMonth,
    SUM(Sales)                AS TotalSales,
    COUNT(OrderID)            AS TotalOrders,
    SUM(Quantity)             AS TotalQuantities
FROM Sales.Orders
GROUP BY DATETRUNC(month, OrderDate);
GO
```

→ *Query the view*

```
SELECT * FROM Sales.V_Monthly_Summary;
```

→ *Drop the view if it exists*

```
IF OBJECT_ID('Sales.V_Monthly_Summary', 'V') IS NOT NULL
    DROP VIEW Sales.V_Monthly_Summary;
GO
```

→ *Re-create with modified logic (fewer columns)*

```
CREATE VIEW Sales.V_Monthly_Summary AS
SELECT
    DATETRUNC(month, OrderDate) AS OrderMonth,
    SUM(Sales)                AS TotalSales,
    COUNT(OrderID) AS TotalOrders
FROM Sales.Orders
GROUP BY DATETRUNC(month, OrderDate);
GO
```

2. Use Case — Hide Complexity

→ *View that joins Orders, Products, Customers, and Employees into one clean view*

```
CREATE VIEW Sales.V_Order_Details AS
SELECT
    o.OrderID,
    o.OrderDate,
    p.Product,
    p.Category,
    COALESCE(c.FirstName, '') + ' ' + COALESCE(c.LastName, '') AS CustomerName,
    c.Country          AS CustomerCountry,
    COALESCE(e.FirstName, '') + ' ' + COALESCE(e.LastName, '') AS SalesName,
    e.Department,
    o.Sales,
    o.Quantity
FROM Sales.Orders AS o
LEFT JOIN Sales.Products AS p ON p.ProductID = o.ProductID
LEFT JOIN Sales.Customers AS c ON c.CustomerID = o.CustomerID
LEFT JOIN Sales.Employees AS e ON e.EmployeeID = o.SalesPersonID;
GO
```

3. Use Case — Data Security

→ *EU Sales Team view — excludes all USA data*

```
CREATE VIEW Sales.V_Order_Details_EU AS
SELECT
    o.OrderID,
    o.OrderDate,
    p.Product,
    p.Category,
    COALESCE(c.FirstName, '') + ' ' + COALESCE(c.LastName, '') AS CustomerName,
    c.Country          AS CustomerCountry,
    COALESCE(e.FirstName, '') + ' ' + COALESCE(e.LastName, '') AS SalesName,
    e.Department,
    o.Sales,
    o.Quantity
FROM Sales.Orders AS o
LEFT JOIN Sales.Products AS p ON p.ProductID = o.ProductID
LEFT JOIN Sales.Customers AS c ON c.CustomerID = o.CustomerID
LEFT JOIN Sales.Employees AS e ON e.EmployeeID = o.SalesPersonID
WHERE c.Country != 'USA';
GO
```